

Governor / Primary Frequency Control

State example: x_g

Inputs: $P_{\text{ref}}, \Delta f$

Parameters: R, T_g, K_i

Droop error:

$$e_g = P_{\text{ref}} - \frac{\Delta f}{R}$$

First-order actuator:

$$T_g \dot{x}_g = e_g - x_g$$

Gate command:

$$y_g = \text{sat}(x_g)$$

P_{ref}
 Δf



y_g